

CS 181 Artificial Intelligence (Fall 2023), Final Exam

Instructions

- Time: 10:30 – 12:10 (100 minutes)
- This exam is closed-book, but you may bring one A4-size cheat sheet. Put all the study materials and electronic devices (except a calculator) into your bag and put your bag in the front, back, or sides of the classroom.
- Two blank pieces of paper are attached, which you can use as scratch paper. Raise your hand if you need more paper.
- For multiple choice questions in Question 2–5:
 - ☐ means you should mark ALL choices that apply;
 - ☐ means you should mark exactly ONE choice;
 - When marking a choice, please fill in the bubble or square COMPLETELY (e.g., ☒ and ☒). Ambiguous answers will receive no points.
 - For each question with ☐ choices, you get half of the points for selecting a non-empty proper subset of the correct answers.

1 Multiple choices (10 pt)

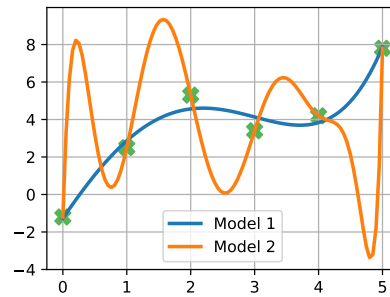
Each question has one or more correct answers. Select all the correct answers. For each question, you get 1 point if you select all the correct answers and nothing else, 0 point if you select one or more wrong answers, and 0.5 point if you select a non-empty proper subset of the correct answers.

1	2	3	4	5
6	7	8	9	10

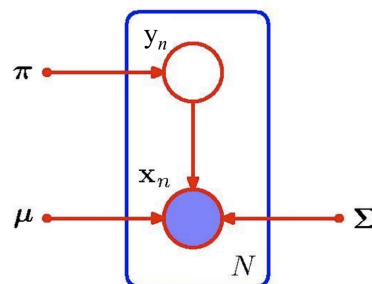
1. Which of the following statement(s) about Markov models, hidden Markov models (HMM), and dynamic Bayes nets (DBN) is/are correct?
 - A. If several states have probability 0 in the initial distribution of a Markov model, then the stationary distribution will never be reached, no matter what the transition model is.
 - B. Considering the formula of the Forward algorithm: $\alpha \times P(e_{t+1} | X_{t+1}) \times \sum_{x_t} P(x_t | e_{1:t}) P(X_{t+1} | x_t)$, the three parts separated by multiplication can be interpreted as “Normalize”, “Update”, and “Predict” from left to right.
 - C. The time complexity of the Viterbi algorithm is $O(|X|^2 T)$ and the space complexity is $O(|X| T)$, where $|X|$ is the number of states and T is the total number of time steps.
 - D. A discrete DBN with 4 triple-state variables in each time step can be converted to a HMM which has 4^3 possible values for each state variable.

2. Which of the following statement(s) about particle filtering is/are correct?
 - A. Likelihood weighting is not proper for HMM because the weight of samples drops quickly.
 - B. In the propagate forward step, each particle is moved by sampling its next position from the transition model $P(X_{t+1}|X_t)$.
 - C. In the observe step, we need to do “count and normalize” to get the weight of each particle.
 - D. In the resample step, each new sample should inherit the weight of the old sample from which it is generated.
3. Suppose we have an MDP with state space of size S and action space of size A . Which of the following statement(s) about value iteration and policy iteration is/are correct?
 - A. Value iteration takes $O(SA^2)$ time for each iteration.
 - B. In value iteration, the values often converge much faster than the policy extracted from the values.
 - C. In policy iteration, we do policy improvement by doing policy extraction from the values computed in policy evaluation.
 - D. Both value iteration and policy iteration can find optimal values.
 - E. None of the above.
4. Which of the following statement(s) about MDP is/are correct?
 - A. MDP could be viewed as randomized search problems, so we can solve them by expectimax search.
 - B. In MDP, the successor state depends on not only the current state, but also the previous states.
 - C. A policy of MDP is a mapping from each state to a reward value.
 - D. A discount factor smaller than one helps the value iteration algorithm converge in general.
 - E. None of the above.
5. Which of the following statement(s) about Q-learning is/are correct?
 - A. To converge to the optimal policy, you need to act optimally during Q-learning.
 - B. In the ϵ -greedy algorithm, agents act randomly with probability $(1 - \epsilon)$.
 - C. Minimizing regret is necessary to learn the optimal policy.
 - D. To converge to the optimal policy, you have to explore enough and eventually make the learning rate small enough.
 - E. None of the above.
6. Which of the following statement(s) about reinforcement learning is/are correct?
 - A. In model-based learning, we first learn the empirical MDP model and then solve the learned MDP.
 - B. In passive reinforcement learning, we do offline planning without taking actions in the world and compute the optimal policy.
 - C. Direct evaluation does not require any knowledge of transition and reward and it utilizes state connections.
 - D. In temporal difference learning, the older samples are given exponentially less weights if the discount factor is less than 1.
 - E. None of the above.
7. A recent paper finds that the performance of LLM (Large Language Model) degrades on test data when repeating training on the training data for additional epochs. Which **ONE** of the following is the most likely reason for this phenomenon?
 - A. The model overfits to the training data.
 - B. The model's capacity is too small to learn the underlying distribution.
 - C. The model's capacity is too large to learn the underlying distribution.
 - D. The model's capacity is too small to learn the underlying distribution.
 - E. The model's capacity is too large to learn the underlying distribution.

- A. The model is overfitting on the test data. Since the model is trained on training data for multiple times, it fits the training data better and better, and it also fits the test data better and better.
 - B. The model is overfitting on the training data. Since the model is trained on training data for multiple times, it fits the training data better and better, but it does not generalize well to the test data.
 - C. The model is overfitting on the training data. Since the model is trained on training data for multiple times, the training data is no longer representative of the true data distribution, so the model does not generalize well to the test data.
 - D. The model is underfitting on the training data. Since the model is trained on training data for multiple times, it does not fit the training data well, and it does not fit the test data well either.
8. Suppose we are doing non-linear regression for a set of data points. The fitting result is shown in the figure below.



- Generally, we prefer Model 1 over Model 2. Which **ONE** of the following is the most likely explanation for this preference?
- A. According to Ockham's razor, we prefer the simplest hypothesis consistent with the data. Model 2 is obviously more complex than Model 1 and overfits the data.
 - B. According to Ockham's razor, we prefer the most complex hypothesis consistent with the data. Model 1 is obviously more complex than Model 2 and fits the data well.
 - C. Model 1 has a lower training error than Model 2, so we prefer Model 1.
 - D. Model 2 fits all the data points perfectly, which is not a good property for a regression model.
9. Which of the following statement(s) about k-means is/are correct?
- A. k-means always converges to global optimal solutions.
 - B. The final clustering results of k-means are usually dependent on the initial cluster centers.
 - C. k-means can only take Euclidean distance as its distance calculation method.
 - D. The point with the minimum squared Euclidean distance to a set of points is their mean.
 - E. None of the above.
10. Consider the sampling process of a Gaussian mixture model (GMM) shown in the figure below.



We choose a multivariate Gaussian component i with probability π_i . Then we generate a sample $\mathbf{x}_n$ from $\mathcal{N}(\mathbf{x}|\mu_i, \Sigma_i)$ with its label $y_n = i$. Which of the following statement(s) about GMM is/are correct?

- A. Given a data sample x_j , we have $\operatorname{argmax}_i P(y_j = i, \mathbf{x}_j) = \operatorname{argmax}_i \pi_i$.
- B. If we are given label $y_j = i$ for each data sample $\mathbf{x}_j$, we can estimate the parameters of GMM by maximizing the likelihood $\prod_j \pi_i \mathcal{N}(\mathbf{x}_j | \mu_i, \Sigma_i)$.
- C. If we do not know label y_j for each data sample $\mathbf{x}_j$, the objective for GMM parameter estimation is to maximize the marginal likelihood $\prod_j \sum_i \pi_i \mathcal{N}(\mathbf{x}_j | \mu_i, \Sigma_i)$.
- D. If we do the generation process many times to produce a lot of data samples, it is most likely true that the larger π_i is, the more data samples are labeled as i .

2 Hidden Markov Models (10 pt)



Intrepid scientists and lovers Katia and Maurice Krafft died in a volcanic explosion doing the very thing that brought them together: unraveling the mysteries of volcanoes by capturing the most explosive imagery ever recorded.

Now they have time-traveled to the present! We can use Hidden Markov Models to assist them in safely exploring the volcano, avoiding risks, and rewriting their tragic ending.

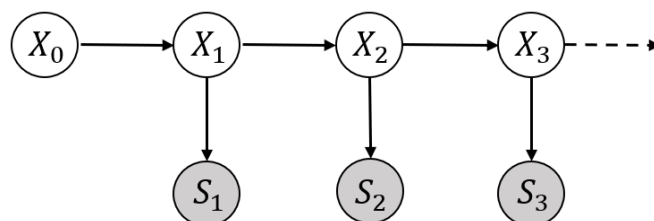
Katia and Maurice Krafft need to go deep into the active volcano to seek for a rare mine. We can only speculate about the existence of rare mines based on the external behavior of the volcano. The behaviour of volcanoes can be divided into three categories: 1. erupting red magma (RED); 2. emitting gray smoke (GREY); 3. being quiet (QUIET).

2.1 Construct Hidden Markov Model (4 pt)

We construct a Hidden Markov Model with the following statistics:

- The prior probability of mine existence (with no observations) is 0.3.
- The probability of mine existence on day t is 0.6 if the mine exists the previous day, and 0.1 if not.
- The probability of the volcano's behaviour being RED/GREY/QUIET is 0.3/0.5/0.2 respectively if the mine exists. On the other hand, the probability of RED/GREY/QUIET is 0.2/0.2/0.6 if the mine doesn't exist.

For state X_t , we have $X_t = 1$ if the mine exists on day t and $X_t = 0$ if not. For evidence S_t on day t , we have $S_t = 0$ if the volcano behaves RED, $S_t = 1$ if the volcano behaves GREY, and $S_t = 2$ if the volcano behaves QUIET.



2.1.1 Complete The Model (2 pt)

Based on the probabilities and model above, fill in the following tables of the initial distribution, transition model and emission model. We have finished part of them for you.

X_0	$P(X_0)$
0	0.7
1	0.3

X_t	X_{t+1}	$P(X_{t+1} X_t)$
0	0	
0	1	
1	0	
1	1	

X_t	S_t	$P(S_t X_t)$
0	0	0.2
0	1	0.2
0	2	0.6
1	0	
1	1	
1	2	

2.1.2 Stationary Distribution (2 pt)

What is the value of $P(X_\infty = 1)$ in the stationary distribution? Choose the correct answer by filling up the circle like ●.

- ☐ 0.05
 ☐ 0.10
 ☐ 0.15
 ☐ 0.20
 ☐ 0.25

2.2 Filtering (3 pt)

We can calculate the posterior distribution over the most recent state given all evidence (i.e., $P(X_t|s_{1:t})$) to do filtering. It helps Katia and Maurice determine whether day t is a good day to seek for mine. Given $s_1 = 1$, $s_2 = 0$, $s_3 = 2$, What is the value of $P(X_3 = 1|s_{1:3})$? Choose the correct answer by filling up the circle like ●.

- ☐ 0.10
 ☐ 0.13
 ☐ 0.16
 ☐ 0.19
 ☐ 0.21

2.3 Most Likely Explanation (3 pt)

Katia and Maurice want to know if they made the right decisions in their past adventures. They sought for the mine in day $t = 2, 3$ and didn't take action in day $t = 0, 1$, given $s_1 = 1$, $s_2 = 0$, $s_3 = 2$. According to most likely explanation (i.e., $\arg \max_{x_{0:t}} P(x_{0:t}|s_{1:t})$), how many right choices have they made in the total four days? Choose the correct answer by filling up the circle like ●.

- ☐ 0
 ☐ 1
 ☐ 2
 ☐ 3
 ☐ 4

3 MDP and RL: Video Pushing (10 pt)

People nowadays spend more and more time browsing videos on their cellphones. For video platforms such as Bilibili and TikTok, it is important to push proper content to different users to maximize the time they spend.

We now formulate this problem as an MDP (markov decision process). There are three state $\{s_1, s_2, s_3\}$, indicating different emotion of user. The actions on these video platforms are simplified to either “change” or “remain”, i.e., whether to change the video content to view next. Different actions may lead to different rewards.

Set the discount factor $\gamma = 0.5$.

3.1 MDP: Value Iteration (5pt)

For this question, suppose we have known the underlying MDP, as shown in the table below.

s	a	s'	$T(s, a, s')$	$R(s, a, s')$
s_1	Remain	s_2	1.0	0.0
s_1	Change	s_3	1.0	-2.0
s_2	Remain	s_1	0.4	-1.0
s_2	Remain	s_3	0.6	2.0
s_2	Change	s_1	0.6	2.0
s_2	Change	s_3	0.4	-1.0
s_3	Remain	s_1	0.6	2.0
s_3	Remain	s_2	0.4	2.0
s_3	Change	s_1	0.4	2.0
s_3	Change	s_2	0.6	0.0

3.1.1 Value iteration (3pt)

Suppose after iteration k of value iteration, we end up with the following values:

$V_k(s_1)$	$V_k(s_2)$	$V_k(s_3)$
0.4	1.4	2.16

Calculate the result after the next iteration:

$V_{k+1}(s_1)$	$V_{k+1}(s_2)$	$V_{k+1}(s_3)$

3.1.2 Policy extraction (2pt)

Suppose value iteration has converged to the following values:

$V^*(s_1)$	$V^*(s_2)$	$V^*(s_3)$
0.8	1.8	2.6

Extract the optimal policy for s_1 :

☐ Remain ☐ Change

Extract the optimal policy for s_2 :

☐ Remain ☐ Change

3.2 Reinforcement learning

In practice, we may not know the underlying MDP, so we need to learn it from data. We now begin with model-based learning.

3.2.1 Model-based learning (2pt)

For the model described above, consider the following samples that the agents have encountered (these samples have been sorted for your convenience):

s	a	s'	r
s_1	Remain	s_2	0.0
s_1	Remain	s_2	0.0
s_1	Remain	s_2	0.0
s_1	Remain	s_3	-10.0
s_1	Remain	s_3	-10.0
s_1	Change	s_3	-8.0
s_1	Change	s_3	-8.0
s_1	Change	s_2	0.0
s_1	Change	s_2	0.0
s_1	Change	s_3	-8.0
s_2	Remain	s_1	-3.0
s_2	Remain	s_1	-3.0
s_2	Remain	s_1	-3.0
s_2	Remain	s_1	-3.0
s_2	Remain	s_3	0.0

s	a	s'	r
s_2	Change	s_1	-10.0
s_2	Change	s_1	-10.0
s_2	Change	s_1	-10.0
s_2	Change	s_1	-10.0
s_2	Change	s_3	0.0
s_3	Remain	s_1	0.0
s_3	Remain	s_2	6.0
s_3	Remain	s_2	6.0
s_3	Remain	s_1	0.0
s_3	Remain	s_1	0.0
s_3	Change	s_2	-8.0
s_3	Change	s_2	-8.0
s_3	Change	s_2	-8.0
s_3	Change	s_1	0.0
s_3	Change	s_2	-8.0

Estimate the following values based on the samples above:

$$T(s_1, \text{Remain}, s_2) = \underline{\hspace{2cm}}, \quad R(s_1, \text{Remain}, s_2) = \underline{\hspace{2cm}}$$

$$T(s_1, \text{Remain}, s_3) = \underline{\hspace{2cm}}, \quad R(s_1, \text{Remain}, s_3) = \underline{\hspace{2cm}}$$

3.2.2 Model-free learning (2pt)

Now, instead of estimating the transition and reward function, we will directly estimate Q values using Q -learning. Assume the step size for Q -learning $\alpha = 0.5$. Our current Q value is as follows:

	s_1	s_2	s_3
Remain	1.5	-0.5	2.7
Change	3.0	-6.0	2.1

Then the agent encounters the following samples

s	a	s'	r
s_1	Change	s_3	8.0
s_3	Change	s_1	0.0

Calculate the updated Q value after these two samples (round to 3 decimal places):

$$Q(s_1, \text{Change}) = \underline{\hspace{2cm}}, \quad Q(s_3, \text{Change}) = \underline{\hspace{2cm}}$$

3.2.3 Exploration v.s. exploitation (1pt)

If the agent selects its action according to probability:

$$\Pr(a|s) = \frac{e^{Q(s,a)/\tau}}{\sum_{a'} e^{Q(s,a')/\tau}}$$

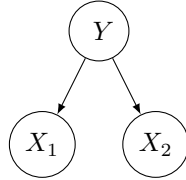
where τ is a parameter that is decreased over time. Then the agent's action is

- ☐ mostly exploration. ☐ mostly exploitation. ☐ mix of both.

4 Supervised Machine Learning (10 pt)

4.1 Naïve Bayes (5 pt)

Answer questions about the following Naïve Bayes model with label $Y \in \{0, 1\}$ and features $X_1, X_2 \in \{0, 1\}$.



When writing your answers, feel free to use fractions like $\frac{2}{3}$.

4.1.1 Maximum Likelihood Estimation (2 pt)

Consider the training data below:

data	1	2	3	4	5	6	7	8	9	10
X_1	1	0	1	0	0	0	1	0	1	1
X_2	1	0	1	0	1	1	1	0	1	1
Y	0	1	1	1	0	0	1	1	1	1

What are the maximum likelihood estimates for the following probabilities?

(1) $P(Y = 0) =$

(2) $P(X_2 = 1|Y = 1) =$

4.1.2 Prediction (1 pt)

Now consider a new data point $X_1 = 1, X_2 = 0$. Use the Naïve Bayes classifier learned in the previous question to determine the posterior probability of Y :

$P(Y = 0|X_1 = 1, X_2 = 0) =$

4.1.3 Laplace Smoothing (1 pt)

Instead of maximum likelihood estimate, we use Laplace Smoothing with strength $k = 1$ to estimate the model with the training data in question 4.1.1. What is the prior $P(Y)$ now?

$P(Y = 0) =$

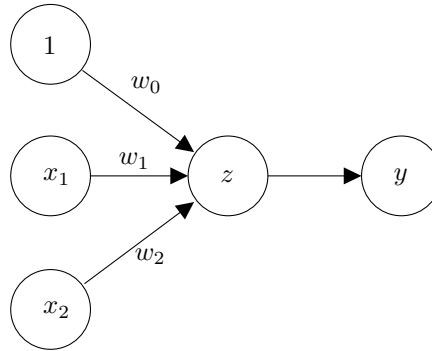
4.1.4 Laplace Smoothing with Large Strength (1 pt)

In practice, Laplace Smoothing with a very large strength k often performs poorly. Which of the following statement(s) is/are possible reason(s) for this phenomenon?

- When k is large, the estimate of $P(Y)$ will be close to a uniform distribution. This is very inaccurate.
- When k is large, the estimate of $P(X_i|Y)$ will be close to a uniform distribution, where $i = 1, 2$. This is very inaccurate.
- When k is large, the estimate of $P(Y)$ will be a sharp distribution with most probabilities close to 0. This is very inaccurate.
- When k is large, the estimate of $P(X_i|Y)$ will be a sharp distribution with most probabilities close to 0, where $i = 1, 2$. This is very inaccurate.

4.2 Perceptron and Logistic Regression (5 pt)

Consider the following computation graph, for which the inputs are two features $x_1, x_2 \in \mathbb{R}$. The output is a binary label $y \in \{-1, +1\}$.



where $z = w_0 + w_1x_1 + w_2x_2$. Suppose we have 2 models: a perceptron and a logistic regression model. The perceptron calculates y as follows:

$$y = \begin{cases} +1 & \text{if } z > 0 \\ -1 & \text{otherwise} \end{cases}$$

The logistic regression model first calculates the probability of $\hat{y} = +1$ as follows:

$$P(\hat{y} = +1|x; w) = \frac{1}{1 + e^{-z}}$$

and then do classification by selecting $y = \arg \max_{\hat{y}} P(\hat{y}|x; w)$.

4.2.1 Classification (2 pt)

What is the output if the weight vector $[w_0 \ w_1 \ w_2] = [1 \ 0 \ 0]$?

- For perceptron, $y = +1$; For logistic regression, $y = +1$.
- For perceptron, $y = +1$; For logistic regression, $y = -1$.
- For perceptron, $y = -1$; For logistic regression, $y = +1$.
- For perceptron, $y = -1$; For logistic regression, $y = -1$.
- Since we do not know the value of x_1 and x_2 , we cannot determine the output.

4.2.2 Update (2 pt)

Suppose we train the perceptron and the logistic regression model respectively on a training data point with input $[x_1 \ x_2] = [1 \ -1]$ and label $y = +1$. How will the weight vector $[w_0 \ w_1 \ w_2]$ be updated if the current weight vector is $[1 \ 0 \ 0]$? Suppose we are doing gradient ascent (maximizing the log likelihood) for logistic regression with learning rate $\alpha = 1$.

	Perceptron			Logistic Regression		
	w_0	w_1	w_2	w_0	w_1	w_2
Increase	☐	☐	☐	☐	☐	☐
Decrease	☐	☐	☐	☐	☐	☐
No change	☐	☐	☐	☐	☐	☐

4.2.3 Gradient Descent (1 pt)

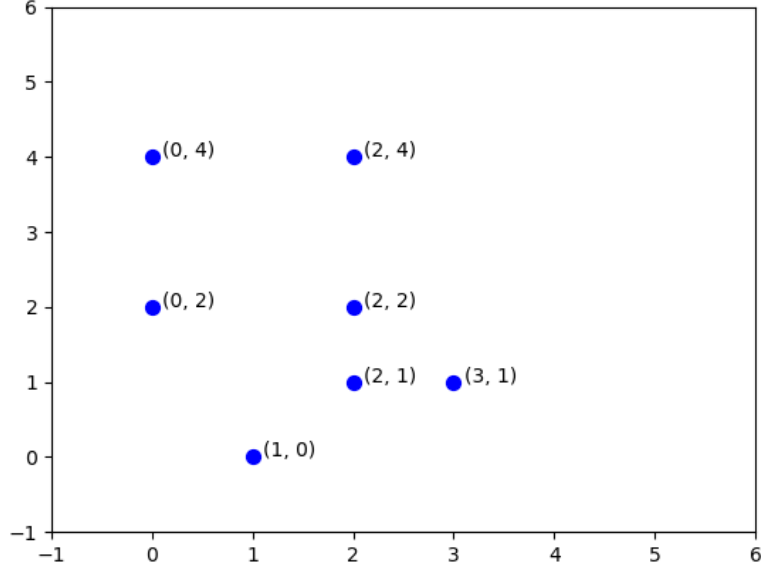
Suppose we have N samples where N is very large. We use gradient ascent to train the logistic regression model. Which **ONE** of the following is the most likely reason for gradient ascent to be very slow, compared with stochastic gradient ascent?

- ☐ For each step we need to compute the gradient on all the N samples, which is very time-consuming.
- ☐ The learning rate α is too small, so that the weight vector does not change much in each step.
- ☐ The learning rate α is too large, so that the weight vector changes too much in each step and the algorithm cannot converge.
- ☐ The loss function has a very flat region, so that gradient ascent converges very slowly.

5 Unsupervised Machine Learning (10pts)

5.1 K-Means (6pts)

Consider the data points in the figure below.



Each data point $s_i \in \mathbb{R}^2$ is represented by a pair (x_i, y_i) . We want to group them into **3** clusters using k-means.

Each cluster center is represented by $c_j^{(t)}$, where $j \in \{0, 1, 2\}$ is the cluster index and $t \in \{0, 1, 2, \dots\}$ is the iteration number. We start from three **initial cluster centers** $c_0^{(0)} = (0, 2)$, $c_1^{(0)} = (2, 1)$, $c_2^{(0)} = (3, 5)$.

We use the Euclidean distance for distance calculation between points, i.e.,

$$\text{dist}(s_i, s_j) = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$$

We run k-means for one iteration in the following questions.

5.1.1 Data Point Assignment (2.5pts)

Assign data points to the closest cluster center. Complete the table below by selecting one of the cluster centers for each data point. We have completed selection for $(0, 2)$ and $(2, 1)$ as an example.

	$(0, 2)$	$(0, 4)$	$(1, 0)$	$(2, 1)$	$(2, 2)$	$(2, 4)$	$(3, 1)$
$c_0^{(0)} = (0, 2)$	■	□	□	□	□	□	□
$c_1^{(0)} = (2, 1)$	□	□	□	■	□	□	□
$c_2^{(0)} = (3, 5)$	□	□	□	□	□	□	□

5.1.2 Update Cluster Center (1.5pts)

Assign each center to the average of its assigned points. Calculate the new cluster centers and fill in the blanks below.

(1) $c_0^{(1)} =$

(2) $c_1^{(1)} =$

(3) $c_2^{(1)} =$

5.1.3 Convergence (2pts)

Does the k-means algorithm converge with $c_0^{(1)}$, $c_1^{(1)}$, $c_2^{(1)}$ as the final cluster centers? (Hint: run one more iteration and check whether the points' assignments will change or not.)

- ☐ Yes, it converges at $c_0^{(1)}$, $c_1^{(1)}$, $c_2^{(1)}$.
- ☐ No, it doesn't converge at $c_0^{(1)}$, $c_1^{(1)}$, $c_2^{(1)}$.

5.2 Expectation Maximization (4pts)

For this section, we will go through a classic problem of Expectation Maximization (EM): the coin flipping problem.

Each coin has two sides: Head (H) and Tail (T). Here, we observe **three** attempts of coin flipping:

attempt 1	HTHTH
attempt 2	HHHHT
attempt 3	THTTH

Suppose we have **two** different coins with **unknown** probabilities of showing Head or Tail. For each attempt, we randomly choose one coin and flip it for 5 times. The results are listed above in order from left to right. For example, the first attempt shows that we flipped one coin 5 times and got Head, Tail, Head, Tail, Head in order.

We define θ_i as the probability of the i_{th} coin showing Head (H) when flipping, define $y_j = i \in \{1, 2\}$ as the label that attempt j is the result of flipping the i_{th} coin, and define $x_j \in \{H, T\}^5$ as the result of attempt j .

Now we use the EM algorithm to estimate θ_1 and θ_2 . We start with a guess that $\theta_1^{(0)} = 0.7$ and $\theta_2^{(0)} = 0.4$. ($\theta_i^{(t)}$ means the estimation of θ_i after t iterations.)

5.2.1 E-Step (2pts)

In E-step, we should compute the label distribution of each attempt j , i.e. $P(y_j|x_j)$. Calculate the distribution of the first attempt and select the correct answer below.

Hint: If one attempt x_k only contains one flip and the result is H, then we have:

$$P(y_k = 1|x_k = H) = \frac{P(x_k = H|y_k = 1)}{P(x_k = H|y_k = 1) + P(x_k = H|y_k = 2)} = \frac{7}{11}$$

The correct value of $P(y_1 = 1|x_1 = (H, T, H, T, H))$ is:

- ☐ 0.573
- ☐ 0.526
- ☐ 0.451

Please also calculate the distribution for the other two attempts and record the results. We need these results for the next question.

5.2.2 M-Step (2pts)

Now we should update the parameter θ in M-step. For the coin flipping setting, we update the new θ_i by:

$$\frac{\text{Expected total number of Head (H) by coin } i}{\text{Expected total number of flips by coin } i}$$

Calculate $\theta_1^{(1)}$ and select the correct answer below. (Hint: For one attempt x_k , the expected number of H by coin i is $P(y_k = i | x_k) \times \{\text{number of H in } x_k\}$.)

The correct value of $\theta_1^{(1)}$ is:

- ☐ 0.713
- ☐ 0.679
- ☐ 0.665

This is a complete iteration of EM for the coin flipping problem. We can keep doing E-step and M-step in turn until convergence to get a final estimation of θ_1 and θ_2 .

CS 181 Artificial Intelligence (Fall 2024), Final Exam

Instructions

- Time: 10:30 – 12:30 (120 minutes)
- This exam is closed-book, but you may bring one A4-size cheat sheet. Put all the study materials and electronic devices (except a calculator) into your bag and put your bag in the front, back, or sides of the classroom.
- Two blank pieces of paper are attached, which you can use as scratch paper. Raise your hand if you need more paper.
- For multiple choice questions in Question 2–5:
 - ☐ means you should mark ALL choices that apply;
 - ☐ means you should mark exactly ONE choice;
 - When marking a choice, please fill in the bubble or square COMPLETELY (e.g., ☒ and ☒). Ambiguous answers will receive no points.
 - For each question with ☐ choices, you get half of the points for selecting a non-empty proper subset of the correct answers.

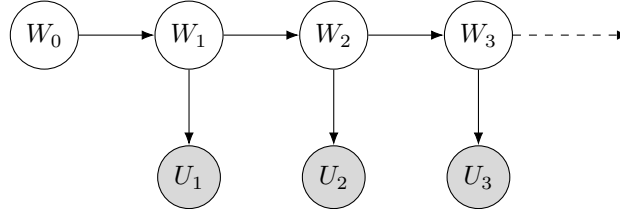
1 Multiple choices (10 pt)

Each question has one or more correct answers. Select all the correct answers. For each question, you get 1 point if you select all the correct answers and nothing else, 0 point if you select one or more wrong answers, and 0.5 point if you select a non-empty proper subset of the correct answers.

1	2	3	4	5
6	7	8	9	10

1. Markov Models are based on the Markov assumption. Consider the first-order Markov models with states $X_1, X_2, X_3, \dots$, which of the following statement(s) is/are correct?
 - A. The Markov assumption states that $P(X_{t+1}|X_{1:t}) = P(X_{t+1}|X_t)$.
 - B. The Markov assumption states that $P(X_{t+1}|X_{1:t}) = P(X_{t+1}|X_{t-1})$.
 - C. Based on the Markov assumption, the past states $X_{1:t-1}$ are independent of the future states $X_{t+1:T}$ given the current state X_t .
 - D. Based on the Markov assumption, the future state X_{t+1} is independent of the current state X_t given the past states $X_{1:t-1}$.

2. Consider the weather Hidden Markov Model (HMM) described in class. The weather state W_t can be either sunny or rainy, and the observation U_t can be either umbrella or no umbrella. Given the observations of $U_1 = \text{umbrella}$, $U_2 = \text{umbrella}$, $U_3 = \text{no umbrella}$, we want to find the most likely weather state sequence $W_{0:3}$. Which ONE of the following algorithms is the most suitable for this task?



- A. Forward algorithm
 - B. Viterbi algorithm
 - C. Gradient descent
 - D. Gibbs sampling
3. Which of the following is NOT a required element in the formal definition of a Markov Decision Process (MDP)?
- A. A set of states $\mathcal{S}$ representing all possible configurations of the environment.
 - B. A set of actions $\mathcal{A}$ representing all possible actions an agent can take.
 - C. A reward function $R(s, a, s')$ that assigns a reward to each transition from state s to state s' via action a .
 - D. An exploration strategy ϵ -greedy that balances exploration and exploitation.
4. Which of the following statement(s) is/are correct?
- A. In policy iteration, we do policy improvement by doing policy extraction from the values computed in policy evaluation.
 - B. The goal of an MDP is to maximize the single-step reward.
 - C. In MDP, the successor state depends on not only the current state, but also the previous states.
 - D. Negative rewards do not exist in MDP.
5. Which of the following statements about the exploration-exploitation tradeoff in Q-learning is *false*?
- A. The ϵ -greedy strategy ensures that the agent continues to explore new actions before ϵ becomes 0, even when it has found a good policy.
 - B. As ϵ decreases over time, the agent will prioritize exploiting the learned Q-values over exploring new actions.
 - C. ϵ -greedy always chooses the optimal action.
 - D. Reducing ϵ too quickly can lead to suboptimal learning because the agent may stop exploring before finding the optimal policy.

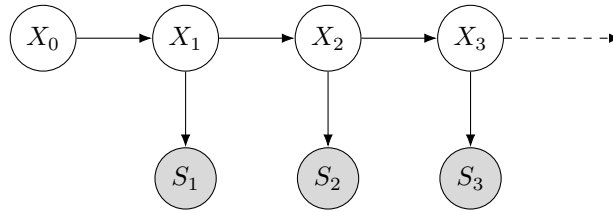
6. Which of the following is necessary for ensuring that Q-learning converges to the optimal Q-values?
 - A. The learning rate α is expected to decrease over time and eventually become sufficiently small.
 - B. The exploration policy should always select the action with the highest Q-value in every state (purely greedy policy).
 - C. The discount factor γ should always be set to 1 to ensure the algorithm prioritizes future rewards equally with immediate rewards.
 - D. The samples to update the Q-function require to be collected by the optimal policy.
7. You are developing two machine learning models to predict house prices based on various features such as size, location, number of bedrooms, and age of the property. After training, you observe that the model *A* performs exceptionally well on the training data but poorly on the validation set. Additionally, a simpler model *B* shows consistently poor performance on both training and validation data. Based on provided information, which of the following statement(s) is/are true?
 - A. The model *A* is overfitting on the training data and it cannot generalize well.
 - B. The model *B* is overfitting on the validation data.
 - C. For model *A*, it is possible to improve the performance on validation data by adding more training data that is representative of the true data distribution.
 - D. For model *B*, it is possible to improve the performance on the same training data by redesigning the model to be more expressive.
8. You are training a logistic regression model using gradient ascent to maximize the log-likelihood function $l(\theta)$ of your data, where θ is the set of model parameters to optimize. Which of the following statement(s) is/are true?
 - A. If the learning rate is set too high, gradient ascent may fail to converge to the maximum.
 - B. Gradient ascent is guaranteed to find the global maximum of $l(\theta)$.
 - C. The output of a logistic regression model is not always in $(0, 1)$ (between 0 and 1).
 - D. Your logistic regression model can also be used for classification.
9. Which of the following statement(s) about self-attention mechanisms in language models is/are correct?
 - A. Self-attention mechanisms try to help the tokens pay more attention to relevant tokens.
 - B. The attention weights are determined by the similarities between the query vectors and the value vectors.
 - C. Causal attention restricts each token to compute attention solely with the tokens before it (include itself) in the sequence.
 - D. Self-attention mechanism only works for sequences of fixed length.
10. Which of the following statement(s) about unsupervised learning is/are correct?
 - A. GMM assumes that data points are generated from a mixture of a finite number of Gaussian distributions.
 - B. Unsupervised learning does not require manual labeling of data.
 - C. The EM algorithm is suitable for solving the parameter estimation problem of Gaussian mixture model (GMM).
 - D. The final result of the K-means algorithm will not be affected by the choice of the initial cluster centers.

2 Sleepy Students (10 pt)

A professor wants to know if students are getting enough sleep. Each day, the professor observes whether the students sleep in class. The professor has the following domain theory:

- The prior probability of getting enough sleep, with no observations, is 0.7.
- The probability of getting enough sleep on night t is 0.8 given that the student got enough sleep the previous night, and 0.3 if not.
- The probability of sleeping in class is 0.1 if the student got enough sleep, and 0.3 if not.

Let X_t denotes the state of getting enough sleep on night t , and S_t denotes the state of sleeping in class on day t . We have the following graphical model:



X_0	$P(X_0)$
0	0.3
1	0.7

X_t	X_{t+1}	$P(X_{t+1} X_t)$
0	0	0.7
0	1	0.3
1	0	0.2
1	1	0.8

X_t	S_t	$P(S_t X_t)$
0	0	0.7
0	1	0.3
1	0	0.9
1	1	0.1

2.1 Filtering (3 pt)

Given the observations $s_1 = 1$, $s_2 = 0$, what is the probability that the student got enough sleep on night 2? That is, what is $P(X_2 = 1|s_{1:2})$?

- ☐ 0.11
 ☐ 0.33
 ☐ 0.55
 ☐ 0.77
 ☐ 0.99

2.2 Particle Filtering (5 pt)

Consider 5 samples in $t = 0$ in the particle filtering algorithm: $X_0^{(1)} = 0$, $X_0^{(2)} = 1$, $X_0^{(3)} = 0$, $X_0^{(4)} = 1$, $X_0^{(5)} = 1$.

2.2.1 Propagate Forward (2 pt)

What are the values of $X_1^{(1)}$, $X_1^{(2)}$, $X_1^{(3)}$, $X_1^{(4)}$, $X_1^{(5)}$ after the propagate forward step?

To generate random samples, use as many values as needed from the table below, which we generated independently and uniformly at random from 0 to 1. Use numbers from left to right. To sample a binary variable W with probability $P(W = 0) = p$, select a value α from the table, and choose $W = 1$ if $\alpha \geq p$ and $W = 0$ otherwise.

We have done the fifth sample for you as an example.

0.123	0.722	0.830	0.476	0.261	0.593
-------	-------	-------	-------	-------	-------

1. $X_1^{(1)}$ ☐ 1 ☐ 0
2. $X_1^{(2)}$ ☐ 1 ☐ 0
3. $X_1^{(3)}$ ☐ 1 ☐ 0
4. $X_1^{(4)}$ ☐ 1 ☐ 0
5. $X_1^{(5)}$ ☒ 1 ☐ 0

2.2.2 Observe (2 pt)

Suppose that the observation is $s_1 = 1$. What is the weight of the sample $X_1^{(5)}$ after the observe step?

- ☐ 0 ☐ 0.1 ☐ 0.286 ☐ 0.3 ☐ 0.714 ☐ 1

2.2.3 Resample (1 pt)

Suppose after observing step of particle filtering at $t = 2$, the particles and its weight are as follow:

Particle	$X_2^{(1)}$	$X_2^{(2)}$	$X_2^{(3)}$	$X_2^{(4)}$	$X_2^{(5)}$
State	0	1	0	1	1
Weight	0.7	0.9	0.7	0.9	0.9

Given the particles and weights after the observe step, what is the weighted sample distribution $P'(X_2)$ used in the resampling step? Select the correct value for $P'(X_2 = 0)$.

- ☐ 0 ☐ 0.134 ☐ 0.341 ☐ 0.413 ☐ 0.431 ☐ 1

2.3 The Argument (2 pt)

The professor prefers the filtering algorithm to the particle filtering algorithm. He also points out that the particle filtering algorithm does not suit the problem of sleepy students. Do you agree with the professor? Please choose the most reasonable explanation from the following options.

- ☐ Yes. Since the state space is very small in this problem, it defeats the motivation of using the particle filtering algorithm.
- ☐ Yes. Since the state space is discrete in this problem, the particle filtering algorithm may produce incorrect results.
- ☐ No. Particle filtering is more efficient than filtering in most cases, including this one.
- ☐ No. Particle filtering could provide more accurate results than filtering in this problem.

3 MDP and RL(10 pt)

3.1 MDP (4 pt)

Consider the MDP with transition model and reward function as given in the table below. Assume the discount factor $\gamma = 1$, i.e., no discounting.

s	a	s'	$T(s, a, s')$	$R(s, a, s')$
A	1	A	0	0
A	1	B	1	0
A	2	A	1	1
A	2	B	0	0
A	3	A	0.5	0
A	3	B	0.5	0

s	a	s'	$T(s, a, s')$	$R(s, a, s')$
B	1	A	0.5	10
B	1	B	0.5	0
B	2	A	1	0
B	2	B	0	0
B	3	A	0.5	2
B	3	B	0.5	4

(a) $V_k(s)$ is the expected value of starting in state s if the game ends in k more time steps, i.e.,

$$V_{k+1}(s) = \max_a \sum_{s'} T(s, a, s') [R(s, a, s') + \gamma V_k(s')] .$$

Let initial $V_0(A) = 0$, $V_0(B) = 0$. Fill in the values for $V_1(A)$ and $V_1(B)$ after one value iteration:

$$V_1(A) = \underline{\hspace{2cm}} \quad , \quad V_1(B) = \underline{\hspace{2cm}}$$

(b) Let $\pi_k^*(s)$ be the optimal action in state s if the game ends in k more time steps, i.e.,

$$\pi_{k+1}^*(s) = \arg \max_a \sum_{s'} T(s, a, s') [R(s, a, s') + \gamma V_k(s')] .$$

Fill in the following tables (fill in 1, 2 or 3):

s	$\pi_1^*(s)$
A	
B	

3.2 Reinforcement learning (6 pt)

Consider an unknown MDP with three states (A, B and C) and two actions ($\leftarrow$ and $\rightarrow$). Suppose the agent chooses actions according to some policy π in the unknown MDP, collecting a dataset consisting of samples (s, a, s', r) representing taking action a in state s resulting in a transition to state s' and a reward of r .

s	a	s'	r
A	$\rightarrow$	B	2
C	$\leftarrow$	B	2
B	$\rightarrow$	C	-2
A	$\rightarrow$	B	4

You may assume a discount factor of $\gamma = 1$.

Assume that all Q -values are initialized to 0, and use a learning rate of $\alpha = \frac{1}{2}$.

(a) Run Q-learning on the above experience table and fill in the following Q -values:

$$Q(A, \rightarrow) = \underline{\hspace{2cm}}, \quad Q(B, \rightarrow) = \underline{\hspace{2cm}}$$

(b) Use the empirical frequency count model-based reinforcement learning method described in lectures to estimate the transition function $\hat{T}(s, a, s')$ and reward function $\hat{R}(s, a, s')$. (Do not use pseudocounts; if a transition is not observed, it has a count of 0.) Write down the following quantities. You may write N/A for undefined quantities.

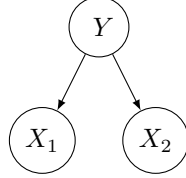
$$\hat{T}(A, \rightarrow, B) = \underline{\hspace{2cm}} \quad \hat{R}(A, \rightarrow, B) = \underline{\hspace{2cm}}$$

$$\hat{T}(B, \rightarrow, A) = \underline{\hspace{2cm}} \quad \hat{R}(B, \rightarrow, A) = \underline{\hspace{2cm}}$$

4 Supervised Machine Learning (10 pt)

4.1 Naïve Bayes (5 pt)

Answer questions about the following Naïve Bayes model with label $Y \in \{0, 1\}$ and features $X_1, X_2 \in \{0, 1\}$.



4.1.1 Play with Naïve Bayes (2 pt)

Consider the training data below:

data	1	2	3	4	5	6	7	8	9	10
X_1	1	1	1	0	1	1	1	0	1	0
X_2	1	0	1	0	1	1	1	0	0	1
Y	0	1	1	1	0	0	1	1	1	1

Maximum Likelihood Estimation (MLE) is a fundamental approach for parameter estimation within statistical models. In real world, Laplace Smoothing is commonly employed. With a smoothing strength of $k = 2$, determine the probabilities for the following scenarios using both MLE (P_{MLE}) and Laplace Smoothing (P_{LAP}). (If necessary, your answers can be in fraction)

(1) $P_{\text{MLE}}(X_2 = 1|Y = 0) =$

(2) $P_{\text{LAP}}(X_2 = 1|Y = 0) =$

4.1.2 Naïve Bayes Prediction (2 pt)

X_1	Y	$P(X_1 Y)$
0	0	θ_1
1	0	$1 - \theta_1$
0	1	0.5
1	1	0.5

Y	$P(Y)$
0	0.5
1	0.5

X_2	Y	$P(X_2 Y)$
0	0	$1 - \theta_2$
1	0	θ_2
0	1	0.5
1	1	0.5

Table 1: CPTs for the Naïve Bayes

Now given the above CPTs, we would like to make predictions on a new sample with $X_1 = 0$ and $X_2 = 1$. What is the probability of $Y = 0$ given $X_1 = 0$ and $X_2 = 1$ ($P(Y = 0|X_1 = 0, X_2 = 1)$)?

- ☐ $\frac{\theta_1(1 - \theta_2)}{\theta_1(1 - \theta_2) + \frac{1}{4}}$
- ☐ $\frac{\theta_2(1 - \theta_1)}{\theta_2(1 - \theta_1) + \frac{1}{4}}$
- ☐ $\frac{\theta_1(1 - \theta_2)}{\theta_1(1 - \theta_2) + \frac{1}{8}}$
- ☐ $\frac{\theta_1\theta_2}{\theta_1\theta_2 + \frac{1}{4}}$

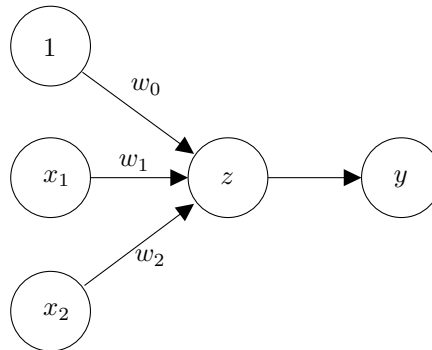
4.1.3 Naive Bayes Classifier (1 pt)

Which of the following statement(s) about the Naive Bayes Classifier is/are **True**?

- ☐ Naive Bayes assumes that all features are dependent given the class.
- ☐ Laplace Smoothing can be used to handle zero probability issues in Naive Bayes.
- ☐ Naive Bayes can be used for multiclass classification problems.
- ☐ The Naive Bayes Classifier always outperforms other classifiers in terms of accuracy.
- ☐ None of the above.

4.2 Perceptron and Logistic Regression (5 pt)

You are working on a binary classification problem using the Perceptron Learning Algorithm. Consider the following computation graph, for which the inputs are two features $x_1, x_2 \in \mathbb{R}$. The output is a binary label $y \in \{-1, +1\}$.



where $z = w_0 + w_1x_1 + w_2x_2$. Suppose we have 2 models: a perceptron and a logistic regression model. The perceptron calculates y as follows:

$$y = \begin{cases} +1 & \text{if } z \geq 0 \\ -1 & \text{otherwise} \end{cases}$$

The logistic regression model first calculates the probability of $y = +1$ as follows:

$$P(y = +1|x; w) = \frac{1}{1 + e^{-z}}$$

and then do classification by selecting $\hat{y} = \arg \max_y P(y|x; w)$. For binary classification, we can have the following prediction rule

$$\hat{y} = \begin{cases} 1 & \text{if } P(y = 1 | x; w) \geq 0.5, \\ -1 & \text{if } P(y = 1 | x; w) < 0.5. \end{cases}$$

The dataset consists of four training examples, each with two features x_1 and x_2 as detailed in Figure 1. We label the positive data (+1) as “+” symbol and the negative data (-1) as “-” symbol. The order of training is from 1 to 4 data point (in red circles), where the data points are (-2, 0), (0, 1), (2, 0), (0, -1).

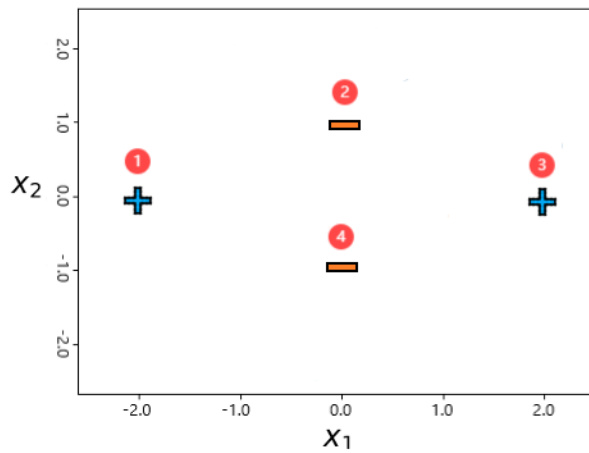


Figure 1: Dataset for perceptron.

4.2.1 Classification (2 pt)

Now we perform binary perceptron update rule (for prediction, please refer to the above) to update the weight vector which is initialized as $[w_0 \ w_1 \ w_2] = [0 \ 0 \ 0]$, where w_0 is the bias term. What is the final weight if we update in the order of $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$ of data points?

4.2.2 Training and prediction (1 pt)

Suppose we train the perceptron on the training data points in Figure 1 in another order ($4 \rightarrow 3 \rightarrow 2 \rightarrow 1$) using features x_1 and x_2 with the same initialized weight. Compared with the resulting weight under the order ($1 \rightarrow 2 \rightarrow 3 \rightarrow 4$) in the above, which of the following statement is correct?

- ☐ They are the same, because the order of training of perceptron will never affect the final weight.
- ☐ They are different, because the order of training of perceptron will possibly affect the final weight.
- ☐ They are the same, though the order of training of perceptron will possibly affect the weight.

4.2.3 Improving Perceptrons (Logistic Regression) (1 pt)

Suppose we train the a logistic regression model on the training data points in Figure 1 using features x_1 and x_2 . Are we able to achieve a consistent 100% accuracy of training (correctly classify all the points)?

- ☐ Yes, because the logistic regression model uses probabilistic decision which is better than perceptron.
- ☐ No, because the dataset is not linear separable.
- ☐ Unknown, because it depends on how well the model is trained.

4.2.4 Feature Exploration (1 pt)

Based on the training data points in Figure 1 using features x_1 and x_2 , you are considering adding one additional feature to the dataset. For each of the following feature transformations, select it if adding it as a third feature (alongside and) would make the two classes linearly separable. Which of the following features should be selected? (Select all that apply.)

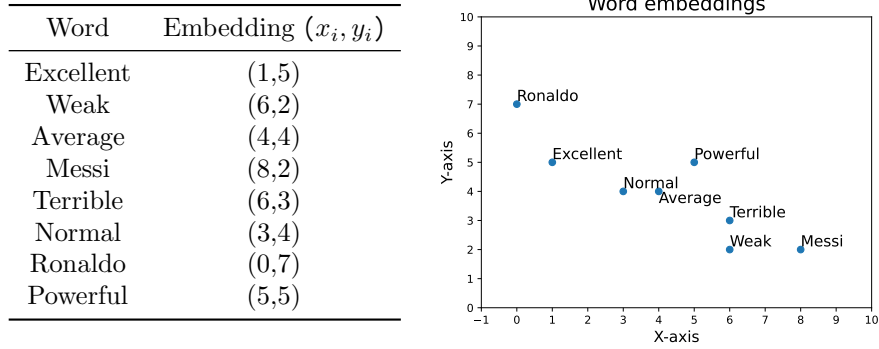
- ☐ $\|x\|_2^2$
- ☐ $\|x\|_2^3$
- ☐ $x_1 + x_2$
- ☐ $2x_1$

For the above, $\|x\|_2$ means the L_2 norm of a vector. For example, if $x = (x_1 \ x_2)$, $\|x\|_2 = \sqrt{x_1^2 + x_2^2}$.

5 Large Language Model (10 pt)

5.1 Word Embeddings (6 pt)

In a large language model, a word embedding is a feature vector that represents each word in a high-dimensional space. Words with similar semantics typically have similar embeddings. As illustrated in the table below, given N word embeddings, our goal is to classify these words using the k-means algorithm.



For simplicity, all word embeddings are represented as two-dimensional vectors $E_i = (x_i, y_i)$ where $x_i, y_i \in \mathbb{R}$.

We use the Euclidean distance for distance calculation between the embedding vectors, i.e.,

$$\text{dist}(E_i, E_j) = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$$

5.1.1 Data Point Assignment (2.5 pt)

The cluster center $c_j^{(t)}$ represents the cluster center of category j after the t iteration. In this problem, we define the number of categories $k = 3$, and the initial three cluster centers are $c_0^{(0)}, c_1^{(0)}, c_2^{(0)} = (1, 5), (3, 4), (5, 5)$.

Assign data points to the closest cluster center given the initial centers.

	(1, 5)	(6, 2)	(4, 4)	(8, 2)	(6, 3)	(3, 4)	(0, 7)	(5, 5)
$c_0^{(0)} = (1, 5)$	■	□	□	□	□	□	□	□
$c_1^{(0)} = (3, 4)$	□	□	□	□	□	■	□	□
$c_2^{(0)} = (5, 5)$	□	□	□	□	□	□	□	■

5.1.2 Update Cluster Center (1.5 pt)

Assign each center to the average of its assigned points. Calculate the new cluster centers and fill in the blanks below.

(1) $c_0^{(1)} =$

(2) $c_1^{(1)} =$

(3) $c_2^{(1)} =$

5.1.3 Convergence (2 pt)

In the K-means algorithm, the choice of initial centers affects convergence speed. Which of the following initial centers can make the K-means algorithm converge faster (convergence time is defined as the number of iterations required)

- ☐ (1, 5), (6, 2), (4, 4)
- ☐ (6, 2), (8, 2), (0, 7)

5.2 Attention is all you need (4 pt)

For this section, we are going to explore the attention mechanism in large language models. The simplified attention weights of a sentence can be calculated by the following formula:

$$A = \text{Softmax}(QK^T, \dim = 1)$$

$A \in \mathbb{R}^{L \times L}$ is the attention weights, $Q \in \mathbb{R}^{L \times D}$ is the query vectors, $K \in \mathbb{R}^{L \times D}$ is the key vectors, where L represents the length of the sentence and D represents the feature dimension. The Softmax function is performed on the last dimension.

5.2.1 Self-Attention (2 pt)

Now we have a sentence $S = \text{"Ronaldo is the best soccer player"}$ which consists of six words. At the same time, we calculated the Q and K of this sentence and get (In this question $D = 2$):

$$S = \begin{bmatrix} \text{Ronaldo} \\ \text{is} \\ \text{the} \\ \text{best} \\ \text{soccer} \\ \text{player} \end{bmatrix}, Q = \begin{bmatrix} 1 & 2 \\ -2 & 1 \\ -3 & 1 \\ 1 & 3 \\ 2 & 2 \\ 2 & 1 \end{bmatrix}, K = \begin{bmatrix} 1 & 1 \\ -3 & 2 \\ -2 & 2 \\ 1 & 2 \\ 2 & 1 \\ 2 & 0 \end{bmatrix}$$

By calculating the attention weights, please answer which word does "Ronaldo" put the most attention weight on?

- ☐ Ronaldo
- ☐ is
- ☐ the
- ☐ best
- ☐ soccer
- ☐ player

5.2.2 Causal Attention (2 pt)

In large language models, causal attention is proposed to make each word only pay attention to the words before it (include itself). The specific implementation method is usually to add a mask when calculating the attention:

$$A = \text{Softmax}(QK^T + M)$$

For the above sentence of length 6, if causal attention is applied, which of the following should the mask M be?

$$\circ \begin{bmatrix} 0, & -\infty, & -\infty, & -\infty, & -\infty, & -\infty \\ -\infty, & 0, & -\infty, & -\infty, & -\infty, & -\infty \\ -\infty, & -\infty, & 0, & -\infty, & -\infty, & -\infty \\ -\infty, & -\infty, & -\infty, & 0, & -\infty, & -\infty \\ -\infty, & -\infty, & -\infty, & -\infty, & 0, & -\infty \\ -\infty, & -\infty, & -\infty, & -\infty, & -\infty, & 0 \end{bmatrix}$$

$$\circ \begin{bmatrix} 0, & -\infty, & -\infty, & -\infty, & -\infty, & -\infty \\ 0, & 0, & -\infty, & -\infty, & -\infty, & -\infty \\ 0, & 0, & 0, & -\infty, & -\infty, & -\infty \\ 0, & 0, & 0, & 0, & -\infty, & -\infty \\ 0, & 0, & 0, & 0, & 0, & -\infty \\ 0, & 0, & 0, & 0, & 0, & 0 \end{bmatrix}$$

$$\circ \begin{bmatrix} 0, & 0, & 0, & 0, & 0, & 0 \\ -\infty, & 0, & 0, & 0, & 0, & 0 \\ -\infty, & -\infty, & 0, & 0, & 0, & 0 \\ -\infty, & -\infty, & -\infty, & 0, & 0, & 0 \\ -\infty, & -\infty, & -\infty, & -\infty, & 0, & 0 \\ -\infty, & -\infty, & -\infty, & -\infty, & -\infty, & 0 \end{bmatrix}$$

CS 181 Artificial Intelligence (Fall 2021), Final Exam

Instructions

- Time: 10:30 – 12:10 (100 minutes)
- This exam is closed-book, but you may bring one A4-size cheat sheet. Put all the study materials and electronic devices (except a calculator) into your bag and put your bag in the front, back, or sides of the classroom.
- Two blank pieces of paper are attached, which you can use as scratch paper. Raise your hand if you need more paper.

1 Multiple choices (10 pt)

Each question has one or more correct answers. Select all the correct answers. For each question, you get 1 point if you select all the correct answers and nothing else, 0 point if you select one or more wrong answers, and 0.5 point if you select a non-empty proper subset of the correct answers.

1	2	3	4	5
6	7	8	9	10

1. Which of the following statements about Markov models, hidden Markov models (HMM), and Dynamic Bayes Nets (DBN) is/are correct?
 - A. For most Markov chains, the stationary distribution is independent of the initial distribution.
 - B. For HMMs, the forward algorithm and the Viterbi algorithm have the same time complexity.
 - C. Every HMM is a DBN, but not every discrete DBN can be represented as a HMM.
 - D. Particle filtering is preferable to exact HMM inference when the state space is large and speed is more important than accuracy.
 - E. None of the above.
2. Which of the following statements are true for MDPs?
 - A. MDPs are non-deterministic search problems.
 - B. When solving MDPs, the value of γ (discount factor) can change the speed of learning, but the final policy that would be learnt remains the same.
 - C. Value Iteration is guaranteed to converge to unique optimal values.
 - D. When running Value Iteration, the values often converge before the policy.
 - E. None of the above.

3. Which of the following is/are not given in Online Learning compared with Offline Planning?
 - A. Set of states S .
 - B. Set of actions A .
 - C. Transition model for states and actions.
 - D. Reward function for every allowed transition between states.
 - E. All of the above.
4. One could simplify the particle filtering algorithm by getting rid of the resampling step and instead keeping weighted particles all the time, with the weight of a particle being the product of all observation probabilities $P(E_i|X_i)$ up to and including the current timestep. Which of the following statements is/are true?
 - A. This will always work as well as standard particle filtering.
 - B. This will generally work less well than standard particle filtering because all the particles will cluster in the most likely part of the state space.
 - C. This will generally work less well than standard particle filtering because most particles will end up in low-likelihood parts of the state space.
 - D. This will generally work less well than standard particle filtering because the number of particles you have will decrease over time.
 - E. None of the above.
5. Which of the following statements about Reinforcement Learning are true?
 - A. Q-Learning is a passive reinforcement learning method.
 - B. Direct Evaluation follows a policy π and makes use of the information about states connections. It is slow because it requires enough samples in order to converge to the true values of states.
 - C. When running Q-Learning with sub-optimal actions, it may still converge to the optimal policy.
 - D. When running Q-Learning, using exploration functions often leads to lower regret than using random exploration.
 - E. None of the above.
6. Which of the following statements about overfitting and underfitting is/are correct?
 - A. Overfitting is more likely to occur when training examples are not representative of the true data distribution.
 - B. Limiting the model expressiveness is an approach to avoiding overfitting.
 - C. Underfitting is more likely to occur when there are insufficient training examples.
 - D. The characteristic of underfitting is that it fits the training data very closely, but does not generalize well.
 - E. None of the above.
7. Which of the following statements about the Naive Bayes classification model is/are correct? There are n feature variables and one label variable. Both feature and label variables are discrete and have a constant domain size.
 - A. The total number of parameters in the Naive Bayes model is linear in n .
 - B. The total number of parameters in the Naive Bayes model is exponential in n .
 - C. The goal of inference in Naive Bayes is to compute the posterior distribution over the label variable.
 - D. An important assumption in Naive Bayes is that the features are independent given the label variable.
 - E. None of the above.

8. Which of the following statements about K-means is/are correct?
 - A. The result of K-means is not influenced by the initial center points.
 - B. K-means converges when there is no longer any point changing the center it is assigned to.
 - C. K-means is an unsupervised learning algorithm.
 - D. K-means is guaranteed to converge to a global optimum.
 - E. None of the above.
9. Which of the following statements about sequence labeling is/are **incorrect**?
 - A. “我骑车差点摔倒，好在我一把把把把住了” This example demonstrates the importance of contextual information in sequence labeling.
 - B. Using a hidden Markov model (HMM) for sequence labeling is better than simply predicting the most frequent label for each word because HMM does not consider relations between adjacent labels.
 - C. Max Entropy Markov Models (MEMM) suffer from the label bias problem.
 - D. Conditional Random Fields (CRF) are undirected graphical models.
 - E. None of the above.
10. Recall that in sequence labeling, HMM inference uses the Viterbi algorithm to find the most likely label sequence of the input sentence. What if we use the Forward algorithm? What does it output? Only ONE answer is correct.
 - A. It outputs the most likely label sequence of the input sentence.
 - B. It outputs the most unlikely label sequence of the input sentence.
 - C. It outputs the most likely word sequence of the label sequence.
 - D. It outputs the posterior distribution over the last label given the input sentence.
 - E. The output of the Forward algorithm does not make sense at all.

Solution:

1. ABD
2. AC
3. CD
4. C
5. CD
6. AB
7. ACD
8. BC
9. B
10. D

2 Hidden Markov Models (10 pt)

Consider a Markov model with a binary state X (i.e., X_t is either 0 or 1). $P(X_0 = 0) = 0.4$ and $P(X_0 = 1) = 0.6$. The transition probabilities are given as follows:

X_t	X_{t+1}	$P(X_{t+1} X_t)$
0	0	0.6
0	1	0.4
1	0	0.3
1	1	0.7

Note: keep three decimal places in your calculation.

(a) Calculate $P(X_1)$. (2pt)

(b) What is the stationary distribution, i.e., $P(X_\infty)$? (2pt)

Now we incorporate binary sensor readings E_t (note that we are starting from E_0). The emission probabilities are given as follows:

X_t	E_t	$P(E_t X_t)$
0	0	0.6
0	1	0.4
1	0	0.7
1	1	0.3

(c) If $E_0 = 1, E_1 = 1$, calculate $P(X_1|E_0, E_1)$. Write down all the steps to get full points. (3pt)

(d) If $E_0 = 1, E_1 = 1, E_2 = 0$, what is the most likely sequence X_0, X_1, X_2 ? Write down all the steps to get full points. (3pt)

Solution:

$$(a) P(X_1 = 1) = \sum_{X_0} P(X_0)P(X_1 = 1|X_0) = 0.4 * 0.4 + 0.6 * 0.7 = 0.58$$

$$P(X_1 = 0) = 1 - 0.58 = 0.42$$

$$(b) P(X_\infty = 1) = 0.3 * P(X_\infty = 1) + 0.4 * P(X_\infty = 0) \rightarrow P(X_\infty = 1) = \frac{4}{7}P(X_\infty = 0)$$

$$P(X_\infty = 0) + P(X_\infty = 1) = 1$$

$$\text{hence, } \frac{11}{7}P(X_\infty = 0) = 1 \rightarrow P(X_\infty = 0) = \frac{7}{11}, P(X_\infty = 1) = \frac{4}{11}$$

(c)

First step (1pt):

$$P(X_0 = 1|E_0 = 1) \propto P(X_0 = 1)P(E_0 = 1|X_0 = 1) = 0.6 * 0.3 = 0.18\alpha$$

$$P(X_0 = 0|E_0 = 1) \propto P(X_0 = 0)P(E_0 = 1|X_0 = 0) = 0.4 * 0.4 = 0.16\alpha$$

The second step (1pt):

$$P(X_t|E_{0:t}) \propto P(E_t|X_t) \sum_{X_{t-1}} P(X_{t-1}|E_{0:t-1})P(X_t|X_{t-1})$$

$$\text{then, } P(X_1 = 1|E_{0:1}) \propto 0.3 * (0.16 * 0.4 + 0.18 * 0.7) = 0.057\alpha$$

$$P(X_1 = 0|E_{0:1}) \propto 0.4 * (0.16 * 0.6 + 0.18 * 0.3) = 0.06\alpha$$

Normalization (1pt):

$$\text{Hence } P(X_2 = 1|E_{0:2}) = 0.057/(0.057 + 0.06) = 0.48717948717949,$$

$$P(X_2 = 0|E_{0:2}) = 0.51282051282051$$

(d)

$$\text{step(0) } m(X_0 = 1|E_0 = 1) \propto P(X_0 = 1)P(E_0 = 1|X_0 = 1) = 0.6 * 0.3 = 0.18\alpha$$

$$m(X_0 = 0|E_0 = 1) \propto P(X_0 = 0)P(E_0 = 1|X_0 = 0) = 0.4 * 0.4 = 0.16\alpha$$

step1: (1pt)

$$m(X_1 = 1|E_{0:1}) = 0.3 * \max(0.16 * 0.4, 0.18 * 0.7) = 0.0378\alpha, (\operatorname{argmax} X_0 = 1)$$

$$m(X_1 = 0|E_{0:1}) = 0.4 * \max(0.16 * 0.6, 0.18 * 0.3) = 0.0384\alpha, (\operatorname{argmax} X_0 = 0)$$

step2: (1pt)

$$m(X_2 = 1|E_{0:2}) = 0.7 * \max(0.0384 * 0.4, 0.0378 * 0.7) = 0.018522\alpha, (\operatorname{argmax} X_1 = 1)$$

$$m(X_2 = 0|E_{0:2}) = 0.6 * \max(0.0384 * 0.6, 0.0378 * 0.3) = 0.013824\alpha, (\operatorname{argmax} X_1 = 0)$$

backtracking (1pt):

$$m(X_2 = 1|E_{0:2}) > m(X_2 = 0|E_{0:2}), \text{ hence, } X_2 = 1$$

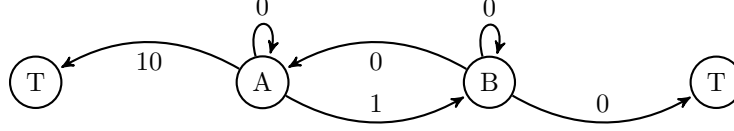
then, $X_1 = 1, X_0 = 1$.

Answer: 1, 1, 1

3 MDP and RL: Left or Right (10 pt)

3.1 MDP (6 pt)

Consider the following MDP.



The state space $\mathcal{S}$ and action space $\mathcal{A}$ are:

$$\begin{aligned}\mathcal{S} &= \{A, B, T\} \\ \mathcal{A} &= \{left, right\}\end{aligned}$$

T denotes the terminal state (the two T states in the figure are the same). When in a terminal state, the agent has no more action to take and gets no more reward. In a non-terminal state (A or B), the agent can only go left or right, but its action only succeeds (going in the intended direction) with probability p_1 . With probability p_2 , it goes in the opposite direction. With probability $1 - p_1 - p_2$, it does not move. In the figure, the numbers on the arrows denote the reward associated with going from one state to another (regardless of the action taken).

For example, if the agent is at state A and takes action *left*:

- with probability p_1 , the next state will be T and the agent will get a reward of 10. The episode is then terminated.
- with probability p_2 , the next state will be B and the reward will be 1.
- with probability $1 - p_1 - p_2$, the next state will still be A and the reward will be 0.

For this problem, the discount factor γ is 1. Let π_p^* be the optimal policy, which may or may not depend on the value of p_1 and p_2 . Let $Q^{\pi_p^*}$ and $V^{\pi_p^*}$ be the corresponding Q and V functions of π_p^* .

(a) If $p_1 = 1, p_2 = 0$, what is π_p^* ?

- ☐ $\pi_p^*(A) = left, \pi_p^*(B) = left$
- ☐ $\pi_p^*(A) = left, \pi_p^*(B) = right$
- ☐ $\pi_p^*(A) = right, \pi_p^*(B) = left$
- ☐ $\pi_p^*(A) = right, \pi_p^*(B) = right$

(b) If $p_1 = 0, p_2 = 1$, what is π_p^* ?

- ☐ $\pi_p^*(A) = left, \pi_p^*(B) = left$
- ☐ $\pi_p^*(A) = left, \pi_p^*(B) = right$
- ☐ $\pi_p^*(A) = right, \pi_p^*(B) = left$
- ☐ $\pi_p^*(A) = right, \pi_p^*(B) = right$

(c) Suppose $\pi_p^*(A) = right$. Which of the following statements must be true? (Select all that apply)

Hint: If $x = y$, then $x \geq y$ and $x \leq y$.

- ☐ $Q^{\pi_p^*}(A, left) \leq Q^{\pi_p^*}(A, right)$
- ☐ $Q^{\pi_p^*}(A, left) \geq Q^{\pi_p^*}(A, right)$

- $\square Q^{\pi_p^*}(A, left) = Q^{\pi_p^*}(A, right)$
- $\square V^{\pi_p^*}(A) \leq Q^{\pi_p^*}(A, left)$
- $\square V^{\pi_p^*}(A) \geq Q^{\pi_p^*}(A, left)$
- $\square V^{\pi_p^*}(A) = Q^{\pi_p^*}(A, left)$
- $\square V^{\pi_p^*}(A) \leq Q^{\pi_p^*}(A, right)$
- $\square V^{\pi_p^*}(A) \geq Q^{\pi_p^*}(A, right)$
- $\square V^{\pi_p^*}(A) = Q^{\pi_p^*}(A, right)$

(d) Assume $p_1 + p_2 = 1, p_1 \geq 0.5$ below:

(i) $V^{\pi_p^*}(B) = \alpha V^{\pi_p^*}(A) + \beta$. Find α and β in terms of p_1 .

• $\alpha =$ _____

• $\beta =$ _____

(ii) $Q^{\pi_p^*}(A, left) = \alpha V^{\pi_p^*}(B) + \beta$. Find α and β in terms of p_1 .

• $\alpha =$ _____

• $\beta =$ _____

(iii) $Q^{\pi_p^*}(A, right) = \alpha V^{\pi_p^*}(B) + \beta$. Find α and β in terms of p_1 .

• $\alpha =$ _____

• $\beta =$ _____

Solution:

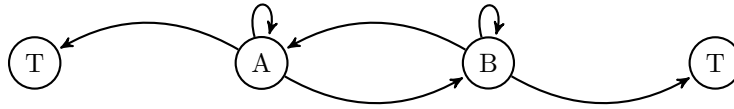
(a) C

(b) B

(c) AEGHI

(d) $\alpha = p_1, \beta = 0$ $\alpha = 1 - p_1, \beta = 9p_1 + 1$ $\alpha = p_1, \beta = 10 - 9p_1$

3.2 Reinforcement Learning (4 pt)



In this part, we are exploring an MDP model with the same structure as above and unknown transition and reward functions. So we will learn the underlying MDP model from experience. Suppose the agent chooses actions based on some policy π and generates the following samples.

t	s_t	a_t	s_{t+1}	r_t
1	B	$left$	B	-1
2	B	$left$	A	2
3	A	$right$	B	1
4	B	$left$	A	2
5	A	$right$	T	5

Assume a discount factor $\gamma = 0.5$ and a learning rate $\alpha = 0.2$.

3.2.1 Model-Based Learning

In model based learning, we first need to estimate the transition model and reward function. Based on the samples above, estimate the following parameters. (Note: p_1 and p_2 have the same definition as in 3.1)

$$p_1 = \underline{\hspace{2cm}} \quad p_2 = \underline{\hspace{2cm}}$$

3.2.2 Model-Free: Q-learning

Assume that all the Q-values are initialized to 0. Apply Q-learning we learnt in class. What are the Q-values learned by running Q-learning with all the transitions shown above? Note that we update Q-values immediately after we receive a sample.

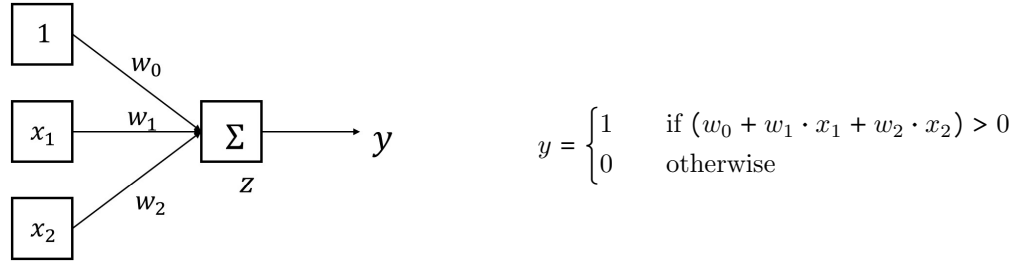
$$Q(B, left) = \underline{\hspace{2cm}} \quad Q(A, right) = \underline{\hspace{2cm}}$$

Solution:

$$p_1 = 0.6, p_2 = 0.2 \quad Q(B, left) = 0.6144, Q(A, right) = 1.1792$$

4 Machine Learning - Perceptron (10 pt)

Consider the following perceptron, for which the inputs are the always-one feature and two binary features $x_1 \in \{0, 1\}$ and $x_2 \in \{0, 1\}$. The output is $y \in \{0, 1\}$.



4.1 Forward & Backward Propagation (2 × 2pt)

- (i) Calculate the output if the weight vector $[w_0 \ w_1 \ w_2] = [1 \ -1 \ 2]$ and input vector $[x_1 \ x_2] = [1 \ 2]$

Your answer: _____

- (i) Say z is an intermediate variable after calculating summation: $z = w_0 + w_1 \cdot x_1 + w_2 \cdot x_2$. What is the derivative of z with respect to x_1 and x_2 respectively?

Your answer: _____; _____

4.2 Weights (2 × 3pt)

- (i) Which of the following choices for the weight vector $[w_0 \ w_1 \ w_2]$ can classify y as $y = (x_1 \text{ OR } x_2)$? Here OR refers to the logical OR operation.

- A. $[1 \ 1 \ 0]$
- B. $[-1.5 \ 1 \ 1]$
- C. $[-2 \ 1 \ 1.5]$
- D. Any weights that satisfy $(-w_1 - w_2) < w_0 < \min(0, -w_1, -w_2)$.
- E. None of the above.

Your answer: _____

- (ii) Which of the following choices for the weight vector $[w_0 \ w_1 \ w_2]$ can classify y as $y = (x_1 \text{ AND } x_2)$? Here AND refers to the logical AND operation.

- A. $[1 \ 1 \ 0]$
- B. $[-1.5 \ 1 \ 1]$
- C. $[-2 \ 1 \ 1.5]$
- D. Any weights that satisfy $(-w_1 - w_2) < w_0 < \min(0, -w_1, -w_2)$.
- E. None of the above.

Your answer: _____

Solution:

Answer: 1

Answer w_1, w_2

Answer: E

Answer: BCD